Conjecture 00000003883: the maximal ideal has F-pure threshold two

lizaixi01, with OpenAI Codex assistance

22 October 2026

1 Statement and the standard definition

The conjecture predicts that the F-pure threshold of (x^a, y^b) equals $\min(a, b)$ when $\gcd(a, b) = 1$ and the characteristic does not divide $\min(a, b)$. Take $a = b = 1$ over the field $\mathbf{F}_2$. These hypotheses hold, but the ideal (x, y) has F-pure threshold 2, not 1.

We work at the origin of the regular polynomial ring $R = \mathbf{F}_2[x, y]$, with homogeneous maximal ideal $\mathfrak{m} = (x, y)$. We use the standard Frobenius definition of the F-pure threshold at that point. For an ideal $\mathfrak{a} \subseteq \mathfrak{m}$, put

$$\nu_{\mathfrak{a}}(2^e) = \max\{r \in \mathbf{N} : \mathfrak{a}^r \not\subseteq \mathfrak{m}^{[2^e]}\}, \qquad \operatorname{fpt}_{\mathfrak{m}}(\mathfrak{a}) = \lim_{e\to\infty} \frac{\nu_{\mathfrak{a}}(2^e)}{2^e}.$$

This definition for non-principal ideals in a polynomial ring is stated explicitly in the introduction of Badilla-Céspedes and León-Cardenal [1], following Takagi–Watanabe [2]. It is the standard F-pure threshold expressed through Frobenius powers. The source's description in terms of test-ideal jumps is brief; here the precise convention is the threshold at the origin. We do not claim to formalize the general equivalence with test-ideal or splitting definitions. We formalize this standard definition and every step of its computation for the exhibited ideal, including existence of the maximum and limit.

2 The computation

Fix $e \geq 0$ and write $q = 2^e$. The iterated Frobenius homomorphism is $F^e(f) = f^q$. Its image ideal on $\mathfrak{m}$ is

$$\mathfrak{m}^{[q]} = (x^q, y^q).$$

A polynomial belongs to this monomial ideal exactly when every monomial in its support has x-exponent at least q or y-exponent at least q. Likewise, a polynomial belongs to $\mathfrak{m}^r$ exactly when all its monomials have total degree at least r.

If $r > 2(q-1)$, every pair of nonnegative exponents of total degree at least r has a component at least q. Thus $\mathfrak{m}^r \subseteq (x^q, y^q)$. Conversely, if $r \leq 2(q-1)$, the monomial $x^{q-1}y^{q-1}$ belongs to $\mathfrak{m}^r$ and does not belong to (x^q, y^q). Therefore, for every $e, r \in \mathbf{N}$,

$$\mathfrak{m}^r \not\subseteq \mathfrak{m}^{[2^e]} \quad \Longleftrightarrow \quad r \leq 2(2^e - 1).$$

The maximum defining ν is consequently attained and is exactly $\nu_{\mathfrak{m}}(2^e) = 2(2^e - 1)$. Hence

$$\operatorname{fpt}_{\mathfrak{m}}(\mathfrak{m}) = \lim_{e\to\infty} (2 - 2^{1-e}) = 2.$$

Since $\gcd(1, 1) = 1$ and $2 \nmid 1$, the uncorrected clause of the conjecture predicts 1. One counterexample to that clause refutes the conjunction; the unspecified correction clause is not needed.

3 Formal objects and verification

`R` is `MvPolynomial (Fin 2) (ZMod 2)`. `maximalIdeal` is the ideal spanned by its two variables. `frobeniusPower` uses `Ideal.map` on Mathlib's actual `iterateFrobenius` ring homomorphism. Theorems `frobeniusPower_span` and `mem_frobeniusPower` establish the generator and support descriptions; these are not assumptions. `power_containment` proves both directions for all e, r. `noncontainment_greatest` uses `IsGreatest` for the actual set of exponents, justifying the `sSup` in `nu`. `threshold_limit` proves convergence in $\mathbf{R}$. `fPureThreshold` uses `limUnder` on that sequence, and `maximalIdeal_fpt` identifies its value using the proved limit. The final existential `counterexample` includes positivity, coprimality, the characteristic condition, actual convergence, and the inequality with the predicted minimum.

Earlier PR 283 attempted a different example but its Lean file only certified disconnected arithmetic. Its claimed universal bound $\mathrm{fpt}(I) \leq 1$ is false for non-principal ideals; the present example itself disproves it. We do not use that bound or assume any comparison with log canonical thresholds. The prior submission's valuation-only discussion is also unnecessary here.

The project pins Lean 4.33.0, Mathlib and all dependencies. Source-bound build, axiom audit and official fresh-kernel replay logs are supplied. No admitted proof, native decision shortcut or new axiom occurs. The bilingual source is fixed at upstream commit `b355e42906e6eaf52bc7e9d005e0ef02d8aa7400`.

OpenAI Codex produced the reasoning, proof and report; a separate supervising Codex agent reviewed the formal chain. This is AI review, not human certification. The formula is classical, not new mathematics.

References

[1] W. Badilla-Céspedes and E. León-Cardenal, *F-Pure Thresholds and F-Volumes of Some Non-Principal Ideals*, Mediterranean Journal of Mathematics 22, 98 (2025), Introduction. `https://doi.org/10.1007/s00009-025-02860-z`.

[2] S. Takagi and K.-i. Watanabe, *On F-pure thresholds*, Journal of Algebra 282 (2004), 278–297. `https://arxiv.org/abs/math/0312486`.